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PAPER

Enhancing image quality in circular-view photoacoustic tomography using randomized detection points

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Abstract

Circular-view (circular scan) photoacoustic computed tomography (PACT) with low-density detection points (DPs) is an efficient, high-speed, and inexpensive modality with numerous (pre-) clinical applications. However, as the number of DPs decreases (decrease in A-lines), some unwanted streak artifacts appear in the reconstructed PACT image. Here, we present an approach to address the mentioned challenge and enhance image contrast. In this method, several low-resolution-images (LRIs) are reconstructed by employing a few DPs' data with randomized locations. These LRIs are used in computing an artifact score matrix (ASM) to identify the location of artifacts. Three numerical (two vasculatures and human brain), two experimental (triangle and complex leaf), and an *in vivo* (a rat brain) studies were conducted to evaluate the efficacy of the proposed method (applying the computed ASM to the final reconstructed PA image). Our findings show that the proposed method outperforms conventional methods and offers better image quality. The signal-to-noise ratio and structural similarity index values of the proposed method are quantitatively 20 dB and 25% better than the conventional method, respectively. Furthermore, compared to the conventional method, the proposed method has an artifact standard deviation that is 50 times lower.

1. Introduction

Photoacoustic (PA) computed tomography (PACT) is a hybrid modality combining both optical and ultrasound imaging [1–5]. The combination of high-contrast optical and deep-penetration ultrasound imaging potentiate the PACT in the diagnosis of a wide range of human diseases and many applications, particularly in breast and brain imaging [6, 7].

In a PACT system, conventional ultrasound detectors acquire the generated PA waves at the medium. The quality of the reconstructed PA image will depend on the coverage of the data acquisition (DAQ) (a full 4π sr solid angle coverage would be ideal) [8]. However, in PACT systems, usually two types of detectors configurations used; (1) planar- and (2) circular-view [3]. In the circular-view, detectors are placed around the medium (2π radian full circular coverage) whereas in planar-view, detectors are placed only on one side (partial coverage) [9, 10]. Compared to planar-view, the generated image with a circular-full-view configuration is of higher quality, as the field of view is larger [11]. For true 3-Dimensional volumetric imaging semi-spherical detectors' array can be used (with larger solid angle coverage) [12].

In the circular-view configuration, the number of detection points (DPs)/ultrasound transducers (USTs) and the radius of the region of interest (ROI) affect the reconstructed image quality [13, 14]. PACT system with low-dense DPs (a few DPs) provides lower image quality compared to high-dense DPs (a large number of DPs). As we reduce the number of DPs, streak artifacts and aliasing appear in the reconstructed image [14]. Typically, to achieve a high-quality image in terms of resolution and contrast, hundreds to thousands of

DPs are required (or number of A-lines collected). If we use a single-element UST, recording many DPs increases the scanning time which prevents real-time imaging. In a system that uses a ring array of detectors, it would be expensive and challenging to manufacture a ring array containing many elements. In addition, to achieve real-time imaging, DAQ card with many parallel channels will be required, making the system very expensive. It would be interesting to have a low-cost PACT system with low-dense DPs (sparse-sampling) that generates high-quality PA images. Recently, to improve image quality, suppress streak artifacts, and remove aliasing in a sparse-sampling PACT system with circular-view configuration, many methods have been reported.

Hu *et al* [15] reported a spatiotemporal anti-aliasing method. They applied a radius-dependent low pass filter and spatial interpolation to raw PA data. This method mitigated streak artifacts and caused blurring in the reconstructed image. In a modified version, they divided the image domain into several subdomains and applied the spatiotemporal method to subdomains [16]. However, compared to the traditional anti-aliasing algorithm, the modified version has more complex filtering and interpolation, which increases the computational cost. In another work, Cai *et al* [17] reported a contamination-tracing back-projection (CTBP) strategy to suppress streak artifacts. They adaptively adjusted the BP weights, to minimize the negative influences of strong absorbers in the reconstructed image. For crowded ROIs, this method may cause significant information loss. Additionally, some tissue dependent parameters and threshold values should be adjusted for each experimental study. Recently, Wang *et al* [18] proposed an anti-streak BP method. They used a threshold value combined with an outlier detection method to find artifacts sources. Then, a weight factor was adjusted to eliminate artifacts. Unfortunately, the selection of the threshold value was tricky and should be adjusted for each experimental study. Applying the coherence-based weighting factors to the reconstructed image suppresses unwanted artifacts. However, some low-intensity/small PA absorbers may also be suppressed [19, 20].

Model- and iterative-based algorithms have been developed to solve image reconstruction problems [21–23]. In ill-conditions problems, regularization methods are developed to improve the quality of reconstructed images [24, 25]. Van de Sompel *et al* [26] deconvolved the recorded PA signals with some deconvolution filters to enhance the image quality. Awasthi *et al* [27] reported an image-guided filter to improve the PACT image obtained from the commonly used Tikhonov regularization method. However, their model matrix was quite large, and this method suffers from heavy computational burdens. Ding *et al* [28] reported a real-time model-based method for tomographic data. Recently, deep learning (DL) methods have shown great success in improving reconstructed image quality [29, 30]. Several DL models have been reported to suppress artifacts and improve the image contrast with a supervised and unsupervised method [31]. Davoudi *et al* [32] proposed a DL method to suppress streak artifacts in circular-view low-dense detector array configuration. Although DL models show great performance in improving the quality of the reconstructed image, these methods usually need large training datasets. Moreover, their performance usually depends on the training datasets, and in different *in vivo* scenarios may be undesirable.

In planar-view configuration scenarios, several methods have been developed to enhance image quality. Short-lag spatial coherence and coherence beamforming-based methods have been developed [33]. Some processing-based methods have been reported to enhance the image resolution of a planar-view configuration (e.g. a signal restoration algorithm [34], sub-aperture adaptive beamformer [35], spatiotemporal coherence weighting [36], delay multiply-and-sum method [37, 38], and spatiotemporal singular value decomposition-based method [39]). Although these methods improve the image quality obtained with planar-view configurations, their performance in circular-view configurations has not been evaluated yet.

The goal of this work is to suppress undesired artifacts and improve the contrast of the generated PA image while avoiding blurring or missing data. Here, we present a processing approach involving randomized DPs selection in a low-density circular-view detector for PACT systems. In our method, we reconstruct an high-resolution-image (HRI) utilizing all DPs' data and hundreds of low-resolution-images (LRIs) by randomly selecting different DPs' data for each image by running a BP technique on PA signals [8, 40]. Because DPs are chosen randomly for each image, streak artifacts occur in random areas while important structural information is reconstructed at certain areas. Then, we compute an artifact score matrix (ASM) to distinguish between reconstructed image locations that refer to artifacts and to the structural information. Finally, we diminish artifacts of the reconstructed image based on the calculated ASM.

In section 2, the materials and methods of our proposed approach are described. In section 3, numerical and experimental results obtained by employing the proposed method are presented. Sections 4 and 5 show the discussions and conclusion.

2. Materials and methods

2.1. Source of artifacts

As a laser source illuminates the PA absorber (PA source) at a location r_m in the ROI, based on the optical absorption coefficient of the PA absorber, a pressure rise $p_0(r_m)$ is induced and propagates across the medium in all directions in the ROI. The discretized form of acoustic pressure at position r_n [position of the n^{th} acoustic detector (AD)] and time t can be expressed as [41]:

$$\hat{p}(r_n,t) = \frac{1}{4\pi c^2}\sum_{m=1}^{M} v_m \frac{p_0(r_m)}{\|r_m - r_n\|} h_{\mathrm{e}}{}'\left(t - \frac{\|r_m - r_n\|}{c}\right), \tag{1}$$

where, c is the speed of sound in the medium, v_m is the volume of the m^{th} excited PA source, M is the total number of excited PA sources, $h_{\mathrm{e}}(t)$ is the electrical impulse response function of the AD, and the prime in h_{e}denotes temporal deviation. A map of the initial pressure rise at the location r'' in ROI to a point at the reconstructed image can be calculated in time-domain by the BP method [8]. The discretized form would be:

$$p_0(r'') \approx \sum_{n=1}^{N} w_n \hat{b}\left(r_n, t = \frac{\|r'' - r_n\|}{c}\right), \tag{2}$$

where, w_n is a weighting factor of the contribution of the nth AD in reconstructing the initial pressure at location r''. The BP term $\hat{b}(r_n,t) = 2\hat{p}(r_n,t) - 2t\frac{\partial \hat{p}(r_n,t)}{\partial t}$. By considering the approximation $\hat{b}(r_n,t) \approx -2t\frac{\partial \hat{p}(r_n,t)}{\partial t}$ [in BP term $\hat{b}(r_n,t)$, the term $2t\frac{\partial \hat{p}(r_n,t)}{\partial t}$ is much larger than $2\hat{p}(r_n,t)$], and substituting (1) in $\widehat{b}(r_n,t)$, $\hat{b}(r_n,t)$ can be rewritten as:

$$\hat{b}(r_n,t) = A\sum_{m=1}^{M} v_m t \frac{p_0(r_m)}{\|r_m - r_n\|} h_{\mathrm{e}}{}''\left(t - \frac{\|r_m - r_n\|}{c}\right), \tag{3}$$

where, A is a constant. Equations (1) and (3) show that for a DP at location r_n, $\hat{b}(r_n,t)$ at each time step t depends on the summation of recorded PA wave values of excited sources that are located in different directions at distance $\frac{t}{c}$ from that DP r_n. Therefore, it is impossible to separate the recorded PA waves from different directions. Since, the ADs are reciprocal, the recorded PA wave of each AD at the time (t) should be back-projected in all directions on a hemispherical surface [in our case a circular arc, as our geometry is in two dimensions (2Ds)] centered at position r_n and it is impossible to map the recorded PA signal to an accurate location. This BP procedure causes streak arc-shaped artifacts in the reconstructed PA image.

According to the summation term in (1), the recorded PA wave of a specific DP associated with a specific distance from that DP would be proportional to the summation of all minor and main PA sources located at the same distance (on the same circular arc) from that specific DP. Minor PA sources refer to low- and very-low-intensity PA sources, and main PA sources refer to general PA sources. The summation term in (1) on all PA sources can be approximated by the summation term in (4) on only the main PA sources,

$$\begin{aligned}\sum_{m=1}^{M} v_m p_0(r_m) &= \sum_{m\in H} v_m p_0(r_m) + \sum_{m\in L} v_m p_0(r_m)\\ &\approx \sum_{m\in H} v_m p_0(r_m),\end{aligned} \tag{4}$$

where, L refers to the location of minor PA sources and H refers to the location of main PA sources. Substituting (4) in (3), would results in:

$\hat{b}(r_n,t) = A\sum_{m\in H} t\frac{v_m p_0(r_m)}{\|r_m - r_n\|} h_{\mathrm{e}}{}''t(t - \frac{\|r_m - r_n\|}{c})$. The back-projected PA signal from the n^{th} DP on a circular arc at a specific time step $t' = \frac{\|r'\|}{c}$ is related to the summation of main PA sources ($m \in H'$) that are located at the same distance$\left(\left\|r_n - r_{m|m\in H'}\right\| = \|r'\|\right)$ from the n^{th} detector. The BP term $\hat{b}(r_n,t')$ for a specific time step t' would be:

$$\hat{b}(r_n,t') = A\sum_{m\in H'} v_m \frac{\|r'\|}{c}\frac{p_0(r_m)}{\|r'\|} = A'\sum_{m\in H'} v_m p_0(r_m). \tag{5}$$

Equation (5) shows that if some main PA sources exist in the ROI, the back-projected ray associated with them will be back-projected not just on those precise spots, but also throughout the entire arc. As a result,

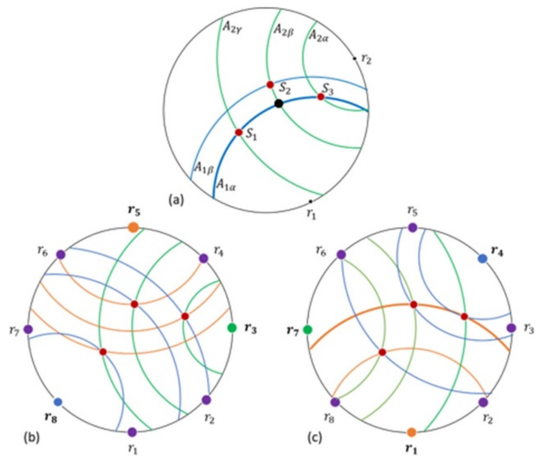

Figure 1. Red dots indicate PA absorbers. (a) Shows back-projected rays referring to PA absorbers from r_1 (blue arcs) and r_2 (green arcs). Black dot shows the interaction of back-projected rays referring to all PA absorbers. (b), (c) show back-projected rays referring to PA absorbers from three DPs with different random locations. Selected detectors for (b), (c) are (r_3, r_5, and r_8) and (r_1, r_4, and r_7), respectively.

some undesirable pressures may appear in the reconstructed image at some locations of the arc that does not refer to those main PA sources, resulting in streak artifacts. Figure 1(a) shows an example of an ROI containing three high-pressure PA absorbers of the same size and intensity (see red dots labeled with $S_1 - S_3$). According to (5), the BP ray from DP r_1 at time step t which refers to the locations of two PA sources, is back-projected on a circular arc with label $A_{1\alpha}$, centered at r_1 and with an intensity value related to the sum of the initial pressure values of those sources (see blue arc with label $A_{1\alpha}$). In the alternative situation, the back-projected ray from same DP referring to another PA source is back-projected on a circular arc with label $A_{1\beta}$, centered at r_1 and with an intensity related to the initial pressure value of that source. Green arcs depict back-projected rays from different time steps for another DP (see green arcs $A_{2\alpha}$, $A_{2\beta}$, and $A_{2\gamma}$ in figure 1(a)). Back-projected rays referring to main sources from different DPs, are thus back-projected on different circular arcs (e.g. for PA source with label S_1 in figure 1(a), parts of the back-projected rays of two detectors r_1, r_2 that refer to source S_1 are back-projected on different arcs $A_{1\alpha}$ and $A_{2\gamma}$).

The back-projected PA signals from all DPs are summed together to reconstruct the initial pressure rise at each site in the ROI $[p_0(r'')]$. By substituting (3) in (2) the map of pressure rise at location r'' becomes as follows:

$$\widehat{p}_0(r'') \approx A\sum_{n=1}^{N} w_n \frac{\|r''-r_n\|}{c}\sum_{m=1}^{M} v_m \frac{p_0(r_m)}{\|r_m-r_n\|} \times h_e''\left(\frac{\|r''-r_n\|}{c}-\frac{\|r_m-r_n\|}{c}\right),. \tag{6}$$

To map each initial pressure in ROI in the reconstructed image, (6) reveals that the back-projected rays of all DPs should be summed. Each pixel in the ROI, have two situations; (1) The pixel refers to a main PA source (r_s''), and (2) the pixel refers to background or a minor PA source (r_B''). If a pixel in the reconstructed image (r''), refers to the main PA source (r_s''), based on (3)–(5), the BP term related to the location of that PA source would be proportional to the initial pressure of that source $\left[\frac{\|r_s''-r_n\|}{c}\sum_{m=1}^{M} v_m \frac{p_0(r_m)}{\|r_m-r_n\|} \propto p_0(r_s'')\right]$. Because the main PA source generates PA waves in all directions, all DPs participate in the reconstruction procedure. As a result, the reconstructed pixel would be proportional to the initial pressure of that main source and the number of ADs that received the generated PA waves, as: $\widehat{p}_0(r_s'') \propto \sum_{n=1}^{N} w_n p_0(r_s'') \propto N p_0(r_s'')$. On the contrary if the reconstructed pixel belongs to background (r_B''), we expect that$\widehat{p}_0(r_B'') \propto 0$. But in practice this does not happen. Based on (5), each PA wave will be back-projected in all directions (on a circular arc), and the main PA sources located on the same arc but in different places may affect the reconstructed pressure of other locations (e.g. see blue and green arcs in figure 1(a)). For example, the location marked with black dot in figure 1(a), is proportional to other PA sources S_1, S_2, and S_3 (see green and blue arcs in figure 1(a)). As a result of substituting (4) in (6), the reconstructed pixel that belongs to the background in the ROI (r_B'') would be proportional to the initial pressure of main PA sources in other locations $(r_m,\ m \in H'')$ that affect the value of $\widehat{p}_0(r_B'')$. Where, H'' refers to main PA sources that affect the reconstructed location (r_B''). Therefore, the reconstructed pixel belongs to a background (r_B'') in the ROI would be as: $\widehat{p_0}\left(r_B''\right) \propto \sum_{m\in H''} w_m p_0\left(r_{m|m\in H''}\right)$. Where, $p_0\left(r_{m|m\in H''}\right)$ is an unwanted initial pressure referring to a PA source in a location that its back-projected signal passes through r_B''. The contrast-to-background (CBR) ratio between a location referring to main a PA source (r_s'') and location referring to the background (r_B'') is calculated by dividing $\widehat{p_0}\left(r_s''\right)$ by $\widehat{p_0}\left(r_B''\right)$ as follows:

$$CBR_{\frac{r_s''}{r_B''}} = \frac{\widehat{p_0}\left(r_s''\right)}{\widehat{p_0}\left(r_B''\right)} = \frac{Np_0\left(r_s''\right)}{\sum_{m\in H''} w_m p_0\left(r_m\right)}. \tag{7}$$

Due to the summation term in $\widehat{p_0}\left(r_B''\right)$, the intensity value of streak artifacts increases as the number of these undesired initial pressures increases, and the CBR ratio decreases. The number of DPs should be increased to enhance the CBR value and minimize the effect of these undesired intensity values. A proposed approach for suppressing artifacts and distinguishing these undesired parts that refer to artifacts and main parts that refer to the location of important structural information is described in the next sub-section.

2.2. Proposed method

In the proposed method, instead of using all the detectors' back-projected rays to reconstruct an HRI, we use only a sub-set of them to reconstruct LRIs in the present approach. In LRI, if a pixel is associated with a main PA source in the ROI, as stated before, all ADs participate in the reconstruction procedure. On the contrary, if the pixel relates to an artifact site, the ADs that cause strike artifacts may or may not be involved in reconstructing that pixel. Although decreasing the number of DPs leads to a decrease in CBR values due to a decrease in the numerator of (7), removing the undesired values caused by main PA sources compensates for the decrease in CBR value and even increases its value due to decreasing the denominator of (7).

In the presented method, we aim to generate some images where the artifacts appear differently in each of them. Considering that the position of the DPs plays a role in the appearance of artifacts (see figure 1(a)), a solution is that the location of the DPs should be different in each image. For the total number of DPs to be constant and not need to obtain data from more DPs, images with a lower resolution than standard can be used. In this way, among all DPs, some of the DPs can participate in reconstructing LRIs.

By reconstructing multiple LRIs with randomly assigned groups of DPs for each LRI, all DPs contribute to reconstructing the pressure rise associated with the PA absorbers. This ensures that the reconstructed pressure rise corresponding to the PA absorbers remains consistent across all LRIs. However, due to the randomized selection of DPs (varying positions of DPs), the artifacts caused by back-projected rays may or may not appear at specific locations in the reconstruction of a particular LRI. Consequently, streak artifacts manifest at varying locations across different LRIs, allowing their effects to be reduced in the final combined reconstruction. For better comprehension, figures 1(b) and (c) illustrate examples of eight DPs and three PA absorbers. We chose two sets of three random DPs ($r_3, r_5,$ and r_8) and ($r_1, r_4,$ and r_7) to reconstruct two LRIs (see figure 1(b) and (c)). In both LRIs, streak artifacts showed up in several places (compare blue, green, and orange arcs in figures 1(b) and (c).

In the proposed method, the reconstructed pressure associated with all DPs for each pixel will be applied to the ASM value of all LRIs related to that pixel. As a result, we anticipate that the intensity value of sites in the reconstructed image that refer to PA absorbers in the ROI would become strengthened, while those that refer to artifacts or background regions would become suppressed.

The detailed workflow diagram of the proposed method is described in the following steps.

Step 1: Generating HRI and reducing the number of DPs to reconstruct LRIs.

After all DPs record PA waves, we reconstruct HRI utilizing all the PA data captured by all DPs. Using fewer DPs to create an LRI may yield a lower-quality image. However, it allows for more flexibility in selecting DPs position for generating LRIs. Consequently, this variation leads to distinct artifacts in each LRI. Therefore, using fewer DPs to generate LRIs allows us to create more LRIs with different artifact positions. However, we should not decrease the number of DPs so much that the main features disappear. As a solution, to reconstruct LRIs, we can reduce the number of DPs by a factor of $N = 2, 3, 4, \ldots$. Until the main features of the LRI begin to fade. To find the optimal number of DPs, we quantify an estimated contrast-to-artifact ratio (CAR), which is a division of the intensities between a PA absorber and a strong artifact in the reconstructed PA image. If the measured CAR is less than one, in a pessimistic condition, the intensity of strong artifacts is approximately the same as the absorbers. In this case, the reconstructed image has exceedingly poor quality in some locations, but the basic features are still distinguishable. Further reducing the number of DPs results in a very low-quality reconstructed image where the main features are extremely difficult to identify, and the reconstructed LRI may not be a suitable candidate for calculating the ASM's values (steps 2 and 3).

The calculation of the CAR value does not require much precision, and the locations in the LRI that refer to an absorber and a strong artifact can be selected approximately.

Step 2: Generating several LRIs by utilizing randomized DPs.

We randomly select several DPs based on the number of DPs determined in Step 1 to reconstruct one LRI. This procedure is repeated to generate multiple LRIs. The number of LRIs to be reconstructed is based on the optimal range discussed in section 4.2, where we find that reconstructing 500–1000 LRIs is sufficient for optimal performance, as any further increases yield insignificant improvements. Since the reconstruction of each LRI is independent, the process can be parallelized and efficiently implemented on a GPU, allowing for rapid processing of multiple LRIs simultaneously.

Step 3: Calculating the artifact score for each pixel.

Then, we calculate the ASM value for each pixel (x, y) in the ROI by quantifying the signal consistency relative to variability across multiple LRIs. The mean pixel value, $\mu(x,y)$, is computed as: $\mu(x,y) = \frac{1}{N}\sum_{n=1}^{N} \mathrm{LRI}_n(x,y)$, where $\mathrm{LRI}_n(x,y)$ is the reconstructed intensity of pixel (x,y) in the n^{th} LRI, and N is the total number of LRIs. The ASM values for each pixel are then defined as: $\mathrm{ASM}(x,y) = \frac{\mu(x,y)^2}{\frac{1}{N}\sum_{n=1}^{N}\mathrm{LRI}_n(x,y)^2}$.

The numerator, $\mu(x,y)^2$, represents the signal consistency, highlighting stable reconstructions across the LRIs, while the denominator, $\frac{1}{N}\sum_{n=1}^{N}\mathrm{LRI}_n(x,y)^2$, accounts for the overall intensity variance, including fluctuations caused by artifacts. By compounding multiple LRIs in this way, a constructive sum of pixel intensities is achieved in locations corresponding to structural information (e.g. PA absorbers), and a destructive sum is achieved in regions associated with artifacts. Since the reconstructed LRIs are identical in areas referring to structural information and vary in regions prone to artifacts, the ASM assigns higher values to structural pixels and lower values to artifacts. Increasing the number of LRIs enhances the suppression of artifacts, as their variability across LRIs is averaged out. As we use more LRIs, the ASM can suppress artifacts better, so we should increase the number of LRIs to reach the optimum state.

Step 4: Applying the ASM to the reconstructed HRI image.

The super-resolved image (SRI) output of the proposed method is obtained by multiplying the ASM's values by the reconstructed HRI. We anticipate suppression of the streak artifacts and strengthening of the locations referred to as the PA absorbers, in the resultant SRI image. The value of the SRI for pixel (x,y) would be: $\mathrm{SRI}(x,y) = \mathrm{ASM}(x,y) \times \mathrm{HRI}(x,y)$.

In the following sub-section, some numerical and experimental (with phantom and *in vivo*) studies were conducted to assay the effectiveness of the proposed method in suppressing artifacts and improving the reconstructed image quality.

2.3. Numerical and experimental studies

2.3.1. Numerical study

The k-Wave MATLAB toolbox [42] was used to generate raw PA data from synthesized numerical phantoms. Three different phantoms were synthesized for the numerical evaluation: (1) simple vasculature phantom (figure 3(a)), (2) complex vasculature phantom (figure 4(a7)), and (3) human brain phantom (figure 4(b6)). The simple vasculature phantom was used to evaluate the quality of reconstructed PA images with different numbers of DPs. The complex vasculature and human brain phantoms were synthesized to assay the performance of the proposed method in suppressing artifacts of the reconstructed image and compare the results with other existing methods. In all the numerical studies 500 LRIs were selected for the proposed method. The speed of sound of the medium was considered 1500 m s^{-1}. The ROI was placed in a square medium (2048 $\times$ 2048 grids) with a grid size of 0.125 mm in both directions. The ROI radius was determined to be 12.5 cm, and ADs were placed on the perimeter of the ROI. After generating raw numerical PA data, for the results to be closer to practical bandwidth-limited ADs, all data were passed through a band-pass filter with Gaussian distribution (center frequency and bandwidth of the filter were selected to be 2.25 MHz and 70%, respectively). To achieve an acceptable image quality, the number of DPs should satisfy the Nyquist criteria, which is: $N = \frac{4\pi R_{\mathrm{ROI}}}{\lambda_c}$, where, λ_c is the higher cut-off frequency of the AD, and R_{ROI} is the radius of the ROI [15]. Based on parameters declared in the numerical study ($\lambda_c = 0.5$ mm, $R_{\mathrm{ROI}} =$ 12.5 cm), to satisfy the Nyquist criterion, at least 3000 DPs are required.

2.3.2. Experimental phantom study

For the experimental evaluation, we used two types of phantoms: (1) a triangular phantom (figure 5(c6)), and (2) a complex leaf phantom (figure 5(d6)). The results of the phantoms were analyzed and compared with different number of DPs and methods. Figure 2(a1) shows the schematic representation of the phantom experimental setup [43]. A Q-switched Nd:YAG laser was used as the excitation source to generate laser pulses of wavelength 532 nm, with a repetition rate of 10 Hz. An acoustic reflector (F102, Olympus NDT) equipped with an unfocused ultrasonic transducer (Olympus NDT, V306-SU) with a center frequency of 2.25 MHz, a nominal bandwidth of 70%, and an active area of 13 mm was utilized to record the generated PA waves in the ROI. Subsequently, the acquired PA signals underwent amplification using a low noise amplifier

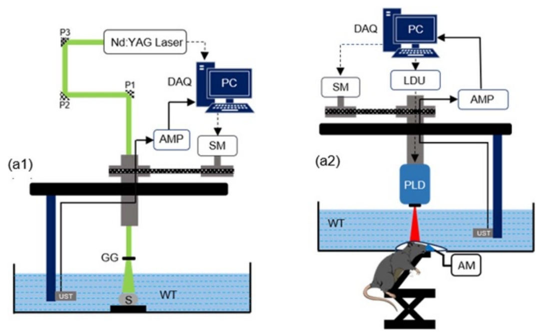

Figure 2. Schematic of the PACT system employed for phantom (a1) and *in vivo* imaging (a2). AMP, amplifier; DAQ, data acquisition card; SM, stepper motor; PC, personal computer; UST, single element ultrasound transducer; GG, ground glass; P1, P2, and P3; uncoated prisms; S, specimen; WT, water tank; AM, anesthesia machine; PLD, pulsed laser diode; LDU, laser driving unit.

(Olympus-NDT, 5072PR) with a 48 dB gain. A DAQ card (GaGe, compuscope 4227) was utilized to record the signals at a 25 MHz sampling frequency. Finally, a desktop computer (IntelXeon, 3.7 GHz 64-bit processor, 16 GB RAM) was used to store the recorded PA signals.

Next, we evaluate the effect of the optimal number of LRIs to converge the proposed method in suppressing artifacts. The results of applying the proposed approach to PA data generated by the triangle target with various numbers of LRIs: 0, 5, 500, and 5 k are shown in figures 5(b1)–(b5), respectively. It should be noted that in other studies 500 LRIs were selected for the proposed method.

2.3.3. In-vivo *study*

We have also assessed our method's performance employing *in vivo* PACT imaging in contrast to other approaches. Sprague-Dawley rat (95 gm) from InVivos Pte. Ltd, Singapore, was used for the *in vivo* imaging. Figure 2(a2) shows the schematic diagram of the *in vivo* PACT imaging system. A mixture of Ketamine (100 mg ml^{-1}) and xylazine (20 mg ml^{-1}) were injected intraperitoneally to anesthetize the rats and the hair on the rat head was removed using the depilatory cream. During imaging, layer of ultrasonic gel was applied on the scalp for ultrasound coupling, and during the imaging process, a steady supply of anesthesia (1.0 l min^{-1} oxygen and 0.75% isoflurane) was provided. A custom-made animal holder equipped with a translational stage was used to position the rat head on the imaging plane. All the *in vivo* experimental procedures were performed in accordance with the Institutional Animal Care and Use Committee (IACUC), Nanyang Technological University, Singapore (Protocol No.: A0331).

The excitation source used for the imaging was a pulsed laser diode (PLD) of ∼816 nm wavelength with a pulse repetition rate of 2 kHz, a pulse width of approximately 107 ns. The laser energy density was kept within the safety limitations stated by the American National Standards Institute (ANSI) [44] at around 0.17 mJ cm^{-2}. For collecting the PA data, an unfocused UST (5 MHz central frequency with 70% nominal bandwidth) equipped with 45° acoustic reflectors (Olympus-NDT, F102) were utilized. More details of the imaging system can be found here [45]. In all reconstructed images, the radius of the ROI was considered ∼3 cm. The required number of DPs to meet the Nyquist criteria for the *in vivo* study was ∼1800. However, we used only 300 and 60 DPs to reconstruct HRI and LRIs, respectively. Two main vasculatures of the rat brain: superior sagittal sinus and transverse sinuses. Also, on the surface of the rat brain, some superficial veins and sigmoid sinuses with low PA absorption coefficient compared to other main vasculatures exist (see the schematic of 3D rat brain vasculatures in figure 5(e6)). Moreover, in the *in vivo* study 500 LRIs were selected for the proposed method.

2.3.4. Compared with existing methods

In this work we present a new method to suppress artifacts and improve image quality. The proposed method is compatible with all reconstruction algorithms and can further enhance the quality of the reconstructed image when combined with DL/CS-based, model-based, and other processing-based approaches. Hence, we did not select the existing DL/CS- and model-based reconstruction methods for comparison with our method. For a fair comparison, our method's experimental and numerical results are compared with those of four existing techniques. (1) Conventional BP (delay-and-sum) algorithm [8]. (2) Coherence weighting factor (CF) [36]. The CF considers the coherence weighting of each pixel in the reconstructed image by calculating the ratio of coherence and incoherence sum of back-projected PA signals referring to each pixel. By multiplying the CF by the reconstructed image, undesired artifacts are suppressed.

(3) Spatiotemporal antialiasing (STAA) method [13]. In the STAA method, at the first stage, authors apply spatial interpolation (Whittaker–Shannon interpolation) and in the second stage, for further suppressing spatial aliasing during spatial samples, they apply a radius-dependent low-pass filter (e.g. third-order low-pass Butterworth and a Sinc filter) to the captured PA signals by the ADs. Unfortunately, although a low-pass filter removes artifacts in the off-center distances in the ROI, however, due to filtering high-frequency components of the captured PA signals, the reconstructed image would be blurred. To avoid blurring and undesired degradation, we implemented the first stage of their method. (4) CTBP method [17]. In this method, authors applied a weighing factor $\left(W^{\mathrm{CB}}\right)$ to all back-projected PA signals for minimizing the influences of contamination sources. The weighing factor W^{CB} consists of three main parameters: 'a', 'w_{win}', and a threshold value. To achieve a high-quality image, these parameters should be carefully selected. These parameters depend on the amount of crowding and intensity of PA absorbers in the ROI. Therefore, different imaging targets required different values of these parameters. Based on their declaration if streak artifacts are strong in the initial reconstructed image, high and low values of 'a' and 'w_{win}' should be selected, respectively. Also, the threshold value should be adopted to achieve the best results. The low value of the threshold results in suppressing more artifacts and low/mid intensity of absorbers which causes missing important data in the reconstructed image. To achieve acceptable results of CTBP method, we set the parameters 'a', 'w_{win}', and the threshold value for the CTBP method within 0.3, 0.15, and 0.7, respectively.

2.3.5. Evaluation metrics

Some image quality metrics are assessed for quantitative evaluation to evaluate performance and compare results achieved with different artifact suppression methods. For numerical studies, the mean of standard deviation (STD), structural similarity index measure (SSIM), contrast-to-noise ratio (CNR), signal-to-noise ratio (SNR), peak-SNR (PSNR) and generalized CNR (gCNR) [46] were measured. In experimental studies, there is no ground truth. Therefore, the mean of STD values of some parts that refer to the background, PSNR, CNR, and SNR values were measured.

3. Numerical and experimental results

3.1. Numerical results

3.1.1. Simple vasculature phantom

In order to determine the optimal number of DPs to reconstruct LRIs, an example containing the numerical results of the simple vasculature phantom with a variety of DP counts (less than 3,000) are shown in figures 3(b1)–(b7). The quality of the reconstructed image degrades by reducing the number of DPs (compare figures 3(b1)–(b6)). The CAR value between a specific absorber and a specific strong artifact (see white arrows in figure 3) decreases to one when the number of DPs is reduced to 66. Although the reconstructed image has exceedingly poor quality, the basic features are a bit distinguishable (figure 3(b6)). Further reducing the number of DPs results in a very low-quality reconstructed image where the other main features are hard to identify (figure 3(b7)).

3.1.2. Complex vasculature phantom

The performance of the proposed method in suppressing artifacts was tested with different numbers of DPs. The reconstructed images with the conventional (DAS)/proposed method with 3000 and 500 DPs are shown in figures 4(a1)–(a4), respectively. Table 1 presents quantitative values of reconstructed images by the standard DAS and the proposed method with various numbers of DPs. The CNR and STD values are calculated for ROIs marked with green and yellow boxes, respectively, and the SSIM values are measured for all the images' pixels. As we decrease the number of DPs in the conventional method, more unwanted artifacts appear in the off-center distances, and the quality of the reconstructed image degrades rapidly. However, the image quality obtained by the proposed method degrades with a slower slope even with 500 DPs (compare white arrows in figures 4(a1)–(a4) and SSIM values in table 1). Quantitatively speaking, by decreasing the number of DPs from 3000 to 500, the SSIM value of the proposed and conventional methods drops $\sim$15% and $\sim$83%, respectively. Moreover, the SSIM and CNR values of the proposed method with 500 DPs are higher than the conventional method with the same number of DPs and almost equal to the conventional method with 3000 DPs. In the case of 3000 DPs, the proposed method improves the CNR value by $\sim$24.5 dB and suppresses the artifacts by $\sim$50-fold in STD.

To compare the proposed method with existing artifacts suppression methods, for the 500 DPs case, the phantom was reconstructed with the STAA and CTBP methods (see figures 4(a5), (a6)). The calculated SNR values for ROIs marked with white boxes are given in table 2. Although the STAA and CTBP methods eliminate artifacts in off-center distances, some other artifacts have remained in the reconstructed image.

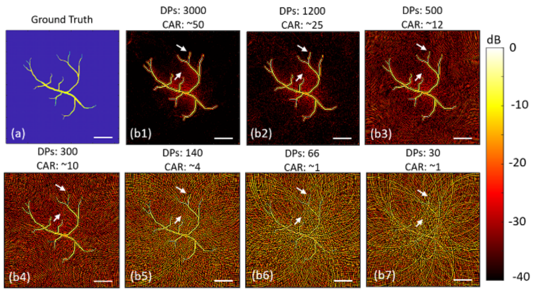

Figure 3. (a) Is the schematic of simple vasculature phantom. (b1)–(b7) are the numerical results of the simple vascular phantom reconstructed with different numbers of DPs. The scale bar is 2 cm.

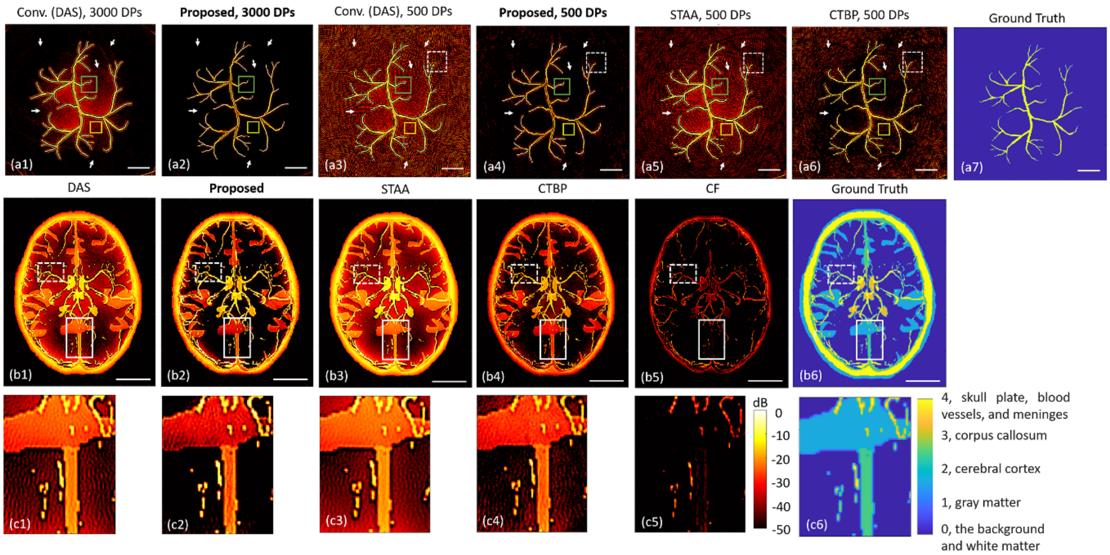

Figure 4. The reconstructed images of complex vasculature (a) and human brain (b), (c) phantoms. (a1)–(a6) are the reconstructed images of the different methods with different number of DPs. (b1)–(b5) are the reconstructed images of the different methods with 500 DPs. The scale bar in (b) is 4 cm. (a7), (b6) show the schematic complex vasculature and 2D human brain phantoms.

Table 1. Quantitative values of conventional and the proposed method with various number of DPS.

DPs	3000 DPs		500 DPs	
Metric	Conv.	Prop.	Conv.	Prop.
SSIM	0.87	0.97	0.41	0.82
CNR	18.5	43.2	14.1	19.8
STD	5×10^{-3}	1×10^{-4}	1×10^{-2}	4×10^{-3}

Table 2. The measured quantitative values of different targets.

Target	Vasculature	Human brain		Triangle targets		Complex leaf phantom	
Metric method	SNR (dB)	PSNR (dB)	gCNR	PSNR (dB)	STDs	PSNR (dB)	STDs
DAS	13.1	20.2	0.62	42	79×10^{-4}	35.7	13×10^{-3}
STAA	14.9	20.8	0.63	43	71×10^{-4}	35.8	13×10^{-3}
CTBP	18.7	25.6	0.71	46	49×10^{-4}	34.5	12×10^{-3}
CF	—	24.1	0.50	—	—	—	—
Prop.	35.2	33.3	0.77	63	62×10^{-4}	50.5	17×10^{-4}

The proposed method suppresses almost all the existing artifacts (compare white arrows in figures 4(a3)–(a6)) and improves the image quality and SNR value by ~22, 20.3 and, 16.5 dB compared to DAS, STAA and, CTBP methods, respectively.

3.1.3. Human brain phantom

The reconstructed images of a 2D human brain phantom obtained with different methods are shown in figures 4(b1)–(b5). The ROIs marked with white boxes are zoomed in figures 4(c1)–(c5). To make the numerical study more realistic, we considered the number of DPs much less than 3 k. We used only 500 DPs to reconstruct HRI with all methods. This 500 DPs is a proper number of DPs for ring array and single elements structures. Moreover, after measuring CAR values based on step 1, we used 75 DPs to reconstruct the LRIs. The gCNR and PSNR values of different methods are calculated for ROIs marked with white and dashed white boxes in figure 4, which are given in TABE II. Based on the results, the STAA and CTBP methods eliminate some artifacts compared to the standard DAS method. However, since the phantom is crowded and contains various PA absorbers with different intensities (e.g. gray matter, cerebral cortex, corpus callosum, extra- and intra-cranial plate, skull plate, and blood vessels), the CTBP and STAA methods did not remove all artifacts without missing some important structural information. Although the CF method suppresses the artifacts to some extent, it also suppresses some low-intensity details which causes missed data effects (compare white boxes in figures 4(b1)–(b6) and gCNR values in table 2). Quantitatively speaking, the PSNR values were measured at 20.2, 20.8, 25.6, 24.1, and 33.3 for DAS, STAA, CTBP, CF, and the proposed method, respectively.

3.2. Experimental results

3.2.1. Triangle target

In section 2.2, we declared that due to the randomized DPs, in the reconstructed image, the streak artifacts' positions will change at each LRI and are dependent on the locations of DPs. To evaluate this assertion, we used 60 DPs to reconstruct the LRIs of the triangle target. An example of eight LRIs with different DPs positions is shown in figures 5(a1)–(a8). The triangle is reconstructed almost well in all LRIs, artifacts (see unwanted arc bows in figure 5(a)) appear in different locations (compare white arrows that refer to unwanted reconstructed arcs in figures 5(a1)–(a8)).

For quantitative evaluation, the STD values of pixels within the white boxes in figures 5(b1)–(b5) are shown as a bar chart in figure 5(b6). Based on the quantitative results (figure 5(b6)) and comparing the reconstructed images (figures 5(b1)–(b5)), by increasing the number of LRIs by more than $\sim$500, the proposed method converges, and the improvement of image quality and STD values are insignificant.

Figures 5(c1)–(c5) show the reconstructed images with 500 DPs obtained with the DAS, proposed, CTBP, CF, and STAA methods, respectively. Since there is no ground truth in the experimental studies, to evaluate the strength of each approach in suppressing artifacts, the PSNR and STD values for the ROIs within white boxes are calculated and given in table 2. STAA and CTBP methods improve image quality and suppress some artifacts. However, some strength artifacts remain. The proposed method suppresses more artifacts than CTBP and STAA approaches without missing important structural information compared to the CF method.

3.2.2. Complex leaf target

Figures 5(d1)–(d5) show the reconstructed images of the complex leaf phantom obtained with the DAS, Proposed, STAA, CF, and CTBP method, respectively. The STD value of ROIs within the white dashed boxes and the PSNR values of the white solid boxes are in table 2. Based on the results, the proposed method suppresses more artifacts than the CTBP and STAA methods. Also, the reconstructed image obtained by the proposed method has higher quality than others (compare figures 5(d1)–(d5)). Since the complex leaf phantom has many PA absorbers and is crowded, the quality improvement of the STAA and CTBP methods is low. The white solid arrows in figures 5(d3) and (d5) show the enhanced contrast areas by the STAA and CTBP methods. Additionally, the green dashed circle in figure 5(d4) represents areas in ROI where the CF technique had missed some crucial information. Quantitatively speaking, the STD values referring to artifacts reach $\sim$0.0017 in the results from the proposed method, which is a $\sim$15-fold improvement compared to other mentioned methods.

3.2.3. In-vivo *rat brain*

In vivo feasibility of the proposed method was demonstrated by performing PA imaging of the healthy rat brain. Figure 5(e) shows the top view reconstructed images of a rat brain obtained with the standard DAS, STAA, CF, CTBP, and the proposed method. Many artifacts appeared in the reconstructed image with the standard DAS method (see arrows in figure 5(e1)). The STAA method suppresses artifacts a bit, and the CTBP and CF methods suppress more artifacts. However, the CTBP and CF methods missed some details of structural information, especially near transverse sinuses and sigmoid sinuses (see dashed white circles in figures 5(e3) and (e4)). The proposed method suppresses undesirable artifacts well and retains important details of structural information.

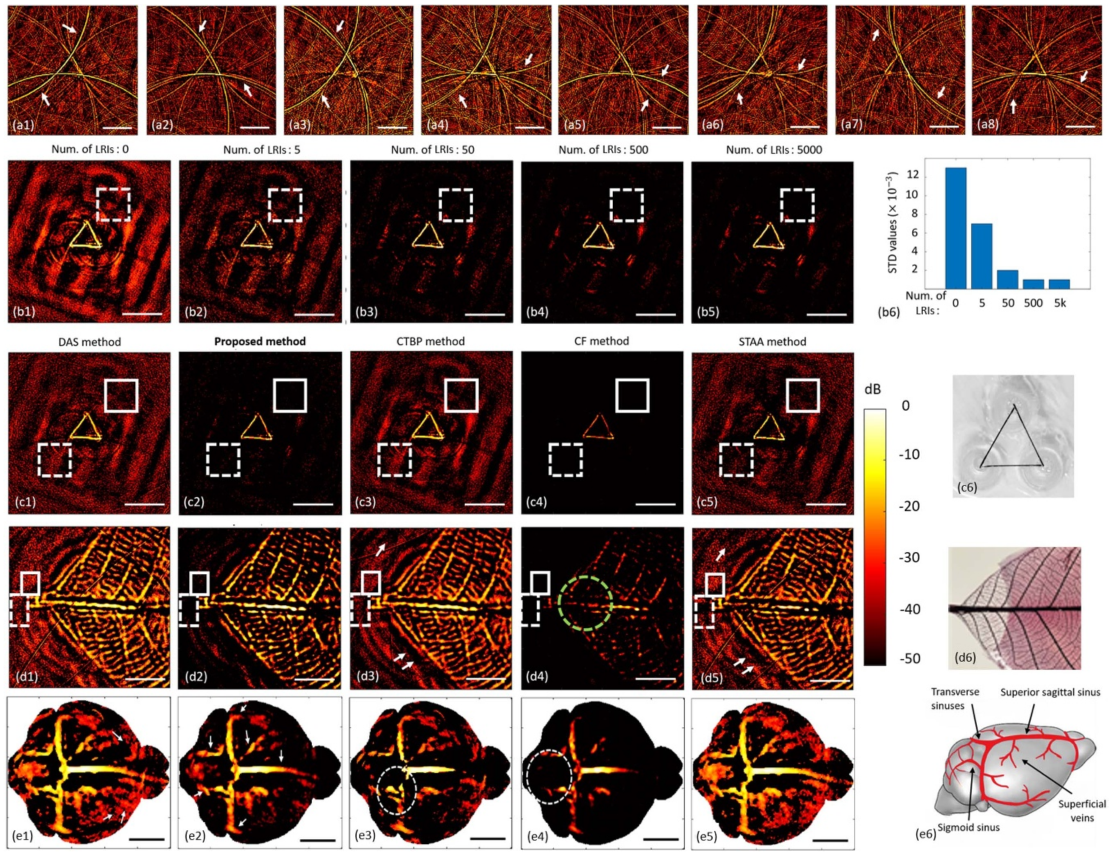

Figure 5. (a1)–(a8) Show an example of reconstructed PACT images of triangle target obtained with 60 DPs in different positions. (b1)–(b5) are the results the triangle target with various numbers of LRIs: 0, 5, 50, 500, and 5 k, respectively. The STD values of pixels within the white boxes in (b1)–(b5) are shown as a bar chart in (b6). (c1/d1–c5/d5) show the reconstructed images of the lab-made triangle/complex leaf targets with 500 DPs obtained with different reconstruction methods. The scale bar in (c), (d) is 1 cm. (e1)–(e5) are cross-sectional reconstructed PACT images of the rat brain obtained with different reconstruction methods. The scale bar in (e) is 1 cm. (c6), (e6) are photographs of the triangle and complex leaf targets, respectively. (e6) is a 3D schematic of rat brain vasculature atlas.

4. Discussions

4.1. Overview

In a circular-view configuration of a PACT system, PA data are usually captured in hundreds of DPs. Most of the time, the number of DPs is much less than the theoretically calculated number to meet the Nyquist criteria. Therefore, many artifacts appear in the reconstructed image degrading the image quality. To address the issue, in this paper, we introduce a novel approach utilizing randomized selection of the DPs. Our method suppresses unwanted artifacts and enhances image contrast in low-density (sparse sampling) DP scenarios. This software-based approach does not necessitate any hardware changes and can be applied to all circular-view PACT systems. The proposed approach could be applicable in clinical and pre-clinical PAT systems such as human brain/breast imaging and whole-body small animal imaging.

4.2. Optimal implementation of the proposed method

According to our studies, determining the optimal number of DPs does not necessitate high precision. Based on the findings, 50–100 DPs are sufficient to generate LRIs. Also, the proposed method reaches its maximum efficiency after reconstructing 500–1000 LRIs, and any gains from utilizing additional LRIs are insignificant. We can implement the reconstruction process of LRIs on GPU kernels because each LRI's reconstruction procedure is independent from the procedure of other LRIs. At first, we launch a CUDA kernel containing 1 k threads and assign each CUDA thread to calculate the randomized locations of DPs and reconstruct one LRI. Then, we launch another CUDA kernel for the ASM calculation in parallel. Finally, we launch a multiplication CUDA kernel to generate the output image. Thus, the proposed approach may be easily and quickly implemented on a GPU, making it suitable for clinical applications. For a GeForce RTX 2060, the entire process to generate one output image (for 500 LRIs and a 512×512 image size) takes approximately 20 ms.

4.3. Comparing the results

First, we conducted numerical studies (e.g. complex vasculature, and human brain phantom) to perform a qualitative and quantitative comparison between the proposed method and different reconstruction methods. These studies revealed that the SNR value of the reconstructed image of complex vasculature obtained with the proposed method is ∼35, DAS; ∼13, STAA; ∼15, and CTBP; ∼19 dB. Although both vasculature phantoms consist of vessels with various diameters, all the absorbers have the same intensity. The human brain phantom consists of absorbers of various sizes and intensities such as white matter, gray matter, cerebral cortex, corpus callosum, skull plate, blood vessels, and meninges (see parula bar in figure 4(c6)). The results show that the STAA method suppresses some artifacts in the white matter areas compared to DAS. The CTBP method suppresses more artifacts than the STAA method. The gray matter and cerebral cortex are more separable than DAS and STAA methods. However, there are still a lot of artifacts in the reconstructed image, particularly in crowded spots such as the locations around vessels and corpus callosum. The CF method suppresses all artifacts but misses some absorbers, such as gray matter and cerebral cortex (compare figures 5(b1)–(b5)). The PSNR value for the proposed method measured ∼13 dB more than the value of the DAS and STAA methods. Also, the gCNR value of the proposed method measured ∼0.77, which is ∼24% improvement compared to DAS and STAA and ∼54% improvement compared to the CF method.

Second, we conducted experimental and *in vivo* studies. The quantitative values for the CF technique were not reported since the CF method omitted certain crucial information. In both phantom studies (triangle and leaf), the PSNR/STD values of the proposed method are higher/lower than other methods. The leaf phantom is more complicated than the triangle phantom, and its STD values are greater, and its PSNR values are lower than those of the triangle target. In the *in vivo* study, the proposed method improves the quality of the reconstructed image. The transverse sinus's morphology along the edges is visible, and the sinuses' boundaries are highly distinct. The proposed method reconstructed important structural information with low intensity, such as some superficial sinuses and sigmoid sinuses (see white arrows in figure 5(e2)), and the CTBP and CF methods missed some of these absorbers (compare tail of transverse sinuses in figures 5(e1)–(e5)). For some applications, such as the detection of intracranial hypotension and cerebral hemorrhaging, this degradation of the venous sinuses represents a major obstacle. Although the standard DAS and STAA methods reconstruct areas that refer to sinuses and veins well, some undesirable artifacts appear more in the reconstructed image than in the proposed method (white arrows in figure 5(e1)). The imaging system used here has a single wavelength excitation. However, the algorithm developed here can be applied to multi-wavelength PA imaging system capable of mapping oxygen saturation we well [47–49].

4.4. Limitations of the proposed method and future work

The proposed method exhibits significant advantages in reducing artifacts and improving image contrast even in low-dense DP scenarios (e.g. about one-tenth of the Nyquist criteria). In very low-dense DP scenarios (e.g. about one-twentieth of the Nyquist criteria) or crowded ROI (e.g. the leaf phantom (figure 5(d2))), the difference number of DPs used to generate LRIs and HRI may not be much. Therefore, the pattern of artifacts may not be much different, and some unwanted distortions and artifacts may remain in the SRI image. In our method, we used the standard DAS method to reconstruct the image and then applied the computed ASM to its output. However, the proposed method is independent of the reconstruction algorithm output. To further improve the result of the proposed approach, the outlined ASM may be combined with alternative weighting techniques, DL/CS-based, and existing reconstruction methods (e.g. model-based, short-lag spatial coherence, and other processing algorithms) that produce fewer artifacts.

5. Conclusion

For circular-view PACT systems, we suggested a data processing method to suppress undesirable artifacts and enhance image quality. In our approach, we identify the source of artifacts by reconstructing multiple LRIs utilizing various DPs that are randomly chosen for each LRI. The results show that our method reconstructs images of higher quality than those reconstructed by conventional methods. Even, the quality of the reconstructed images obtained with our method in low-dense DP scenarios is higher than those obtained with the conventional approach, even in high-dense DP scenarios. Quantitatively, our method increases CNR, PSNR, and gCNR values by up to 25 dB, 20 dB, and 25%, respectively, while suppressing artifact STD values by up to 50-fold.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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